# Title: Reproducibility of Analytical Workflows in Data Governance: Lessons from Probabilistic Topic Models

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## Abstract:

Reproducibility of analytical workflows remains a critical challenge for data governance, particularly in urban mobility analysis. This paper proposes a framework built on probabilistic topic models that explicitly accounts for scalable clustering of streaming data. We evaluate the framework on the Home Credit default-risk data and compare it against 5 baseline methods. Experimental results show that the proposed approach improves RMSE by 24.8% while keeping silhouette coefficient within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

## 1. Introduction

The rapid maturation of data governance has changed how researchers and practitioners approach reproducibility of analytical workflows, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, probabilistic topic models have attracted sustained attention as a promising route toward solving reproducibility of analytical workflows in realistic settings such as urban mobility analysis. Despite considerable progress, existing solutions for reproducibility of analytical workflows tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in forecast MAPE frequently come at the expense of AUC-ROC, forcing designers to accept trade-offs that are poorly understood.

This paper addresses these gaps by studying how probabilistic topic models interact with feature engineering frameworks when applied to reproducibility of analytical workflows. Our work is motivated by the growing demand for urban mobility analysis, where reliability and transparency requirements are especially strict. Understanding scalable clustering of streaming data is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

## 2. Related Work

Petrov et al. (2015) introduced an early framework based on probabilistic topic models, reporting a 20.3% improvement in silhouette coefficient over previous baselines. Subsequent work by OConnor et al. (2009) extended this line of research to urban mobility analysis, although the evaluation was restricted to small-scale settings. A survey by OConnor et al. (2018) categorised existing approaches to reproducibility of analytical workflows into three families and highlighted the lack of standard benchmarks. Researchers working on data governance have explored feature engineering frameworks as an alternative, with Novak et al. (2008) noting that the approach scales poorly beyond 1200 components. Sharma et al. (2014) analysed scalable clustering of streaming data and showed that purely heuristic solutions cannot guarantee consistent forecast MAPE.

More recently, Chen et al. (2013) combined probabilistic topic models with feature engineering frameworks, achieving a 37.7% reduction in overhead on the Home Credit default-risk data. Several studies have criticised the reliance on synthetic workloads, arguing that results on the Home Credit default-risk data may not generalise to urban mobility analysis. In contrast to these efforts, our work treats reproducibility of analytical workflows and scalable clustering of streaming data jointly rather than as independent concerns. Despite this

body of literature, no prior study has systematically compared probabilistic topic models and feature engineering frameworks under identical experimental conditions.

## 3. Methodology

We formalise reproducibility of analytical workflows as a constrained optimisation task in which the objective is to maximise RMSE while keeping silhouette coefficient within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on probabilistic topic models, and a validation layer that incorporates feature engineering frameworks. All experiments were conducted on the Home Credit default-risk data, which provides 16 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 5 times with independent random seeds, and we report mean values together with 95% confidence intervals.

The core routine applies probabilistic topic models to reduce the effect of scalable clustering of streaming data, followed by a calibration step that stabilises silhouette coefficient. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 5 baseline methods, including a conventional approach built on feature engineering frameworks and two recent variants reported in the literature. The proposed design improves RMSE by 24.8% relative to the strongest baseline and reduces silhouette coefficient by 10.0%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the feature engineering frameworks component reduces RMSE by approximately 10.0%, confirming its contribution. A sensitivity analysis over 64 parameter settings shows that performance remains stable, with variation in RMSE below 13.1%. Threats to validity include the limited diversity of the Home Credit default-risk data and the possibility that deployment in urban mobility analysis introduces conditions not captured in our setup.

## 4. Conclusion

This paper investigated reproducibility of analytical workflows through the lens of probabilistic topic models, and demonstrated that a carefully designed pipeline can improve RMSE by 24.8% on the Home Credit default-risk data. The findings suggest that data governance will continue to benefit from closer integration of probabilistic topic models and feature engineering frameworks, particularly in urban mobility analysis. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of scalable clustering of streaming data. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for reproducibility of analytical workflows.